

# Majid Ghasemi

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## Research interests

Social Reinforcement Learning, Multi-agent Reinforcement Learning, Machine Ethics, AI Alignment, and AI Ethics

## Education

### University of Waterloo

Waterloo, ON, Canada

**Ph.D.** in Electrical & Computer Engineering

Sep 2025 – Present

Supervised by Mark Crowley

Research area: *Ethical decision making in multi-agent systems and how social learning can be a way to achieve it*

### Wilfrid Laurier University (WLU)

Waterloo, ON, Canada

**MASc** in Applied Computing

Sep 2023 – May 2025

Supervised by Dariush Ebrahimi. *GPA: A+.*

Thesis title: *Reinforcement Learning for Scheduling and Routing: Electric Vehicles Charging and Patrol Scheduling*

### Imam Khomeini Intl. University (IKIU)

Qazvin, Iran

**B.Sc.** in Software Engineering

Sep 2017 – March 2022

Supervised by Mahdi Bahaghighat. *GPA: A-.*

## Publications

### Citation count prediction based on Google Scholar profiles and Clarivate's journal citation reports

Mahdi Bahaghighat, Leila Akbari, *Majid Ghasemi*, and Qin Xin.

*Information Research journal*

### Toward Virtuous Reinforcement Learning: A Critique and Roadmap

*Majid Ghasemi*, and Mark Crowley.

*Machine Ethics: from formal methods to emergent machine ethics workshop at the AAAI conference*

### Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response

*Majid Ghasemi*, and Dariush Ebrahimi.

*IECON 2025 – 51st Annual Conference of the IEEE Industrial Electronics Society.*

### Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban Safety

*Majid Ghasemi, Ebrahim Sarkhouh, Fadi Alzhouri, and Dariush Ebrahimi.*  
*2025 IEEE 34th International Symposium on Industrial Electronics (ISIE).*

**Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging**

*Majid Ghasemi, Fadi Alzhouri, and Dariush Ebrahimi.*  
*2025 21st International Conference on the Design of Reliable Communication Networks (DRCN).*

**Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Challenges**

*Majid Ghasemi, Amirhossein Mousavi, and Dariush Ebrahimi.*  
*Preprint on Arxiv.*

**Introduction to Reinforcement Learning**

*Majid Ghasemi, and Dariush Ebrahimi.*  
*Preprint on Arxiv.*

**Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra's Algorithm**

Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, *Majid Ghasemi, Fadi Alzhouri, and Dariush Ebrahimi.*  
*2024 International Conference on the Dynamics of Information Systems.*

**A high-accuracy phishing website detection method based on machine learning**

Mahdi Bahaghighat, *Majid Ghasemi, and Figen Ozen.*  
*Journal of Information Security and Applications 77.*

Honors and  
scholarships

|                                                       |             |
|-------------------------------------------------------|-------------|
| Winner of the Academic Excellence Gold Medal (WLU)    | 2025        |
| GSRDA Award (University of Waterloo)                  | 2026        |
| Int'l Doctoral Student Award (University of Waterloo) | 2025 – 2029 |
| Graduate Scholarship (WLU)                            | 2023 – 2025 |
| Graduate Fellowship (WLU)                             | 2023 – 2025 |
| Differential Tuition Fee Waiver (WLU)                 | 2023 – 2025 |
| Excellence Students Tuition Fee Waiver (IKIU)         | 2017–2022   |

Research experience

**UWECML/CompThink lab**

Supervisor: Prof. Mark Crowley (University of Waterloo) Sep 2025 – Present  
Developing socially informed and ethically grounded approaches to multi-agent reinforcement learning.

### **Edge Computing lab**

Supervisor: Dr. Dariush Ebrahimi (WLU)

Sep 2023 – May 2025

Conducted research on reinforcement learning for intelligent transportation systems, focusing on EV fleet management and routing optimization.

### **AIST lab**

Supervisor: Dr. Mahdi Bahaghighat (IKIU)

Sep Jan. 2021 – March 2022

Researched on phishing website detection with machine learning.

## Teaching experience

### **Teaching Assistant, IKIU, WLU, and UW**

2023-Present

Courses including AI, Data Mining, Algorithm Design and Analysis, and Discrete Mathematics

## Industry experience

**Kisoji Biotechnology Inc.**, Machine Learning Department    Waterloo, ON, Canada

AI Research Intern

May 2026 – Present

Currently building the Expressibility Engine — a 3-module system for antibody expression optimization: domain-adaptive prediction with Bayesian project embeddings, RL-guided experimental design (GRPO + DPP diversity), and sparse graph-based causal driver discovery. Reduces wet-lab rounds via active learning while providing interpretable guidance for antibody engineers.

## Service to the Community

### **(R) Stands for Reviewer, (PC) stands for Program Committee**

(PC) AAAI/ACM Conference on AI, Ethics, and Philosophy (AIES) 2026

(R) Reinforcement Learning in Big Worlds at Reinforcement Learning Conference (RLC) 2026

(R) PhilML Workshop at ICML 2026

(R) Pluralistic Alignment Workshop at ICML 2026

(R) Post-AGI Science and Society (P-AGI) Workshop at ICLR 2026

(R) IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)

(R) ACM Transactions on Intelligent Systems and Technology

## Grants

### **HU University of Applied Sciences Utrecht**

Aug. 2022 – Aug. 2023

Recipient of a grant to conduct research on Woman, Life, Freedom movement in Iran with computational social science and machine learning approaches.